Tensor computations

Improve the worst-case approximation factor for prescribed tensor-train ranks

Difficulty: extreme **Importance:** interesting to the community

Status: open; explicit source updated 2026-08-20.

Problem statement

Let $d \geq 3$, $n_1, \ldots, n_d \geq 2$, and positive integers $r_1, \ldots, r_{d-1}$ be given. Let $S_r \subset \mathbb{R}^{n_1 \times \cdots \times n_d}$ consist of tensors whose unfolding separating modes $1, \ldots, j$ from modes $j+1, \ldots, d$ has rank at most r_j, for every j. For a dense input tensor A, put

$$E_*(A, r) = \min_{Y \in S_r} \|A - Y\|_F^2.$$

Find a polynomial-time algorithm returning $X \in S_r$ with

$$\|A - X\|_F^2 < (d-1) E_*(A, r) \qquad \text{whenever } E_*(A, r) > 0,$$

and exact reconstruction when $E_*(A, r) = 0$; alternatively, establish a complexity obstruction to such a guarantee. Ranks may not be increased. Polynomial time is measured in the dense input size and rank parameters, in the idealized arithmetic/SVD model used for TT-SVD; a bit-complexity formulation must additionally specify precision and output tolerances.

This is the tensor-train case of the published tree-network question. The positive-error qualification corrects the impossible strict inequality at zero optimum. It asks for a uniformly valid algorithm, not an empirical improvement or a guarantee restricted to particular input tensors.

References

I. V. Oseledets, *Tensor-Train Decomposition*, *SIAM Journal on Scientific Computing* 33 (2011), 2295–2317, Theorem 2.2 and Corollary 2.4. Amsel et al., workshop report, §6.1, Problem 6.1. Matthew Fahrbach and Mehrdad Ghadiri, *A Tight Lower Bound for the Approximation Guarantee of Higher-Order Singular Value Decomposition*, Theorems 1.2–1.3, concerns tightness of specific Tucker algorithms rather than a lower bound against all tensor-train algorithms.

Status check

Searches included *"tensor train" "approximation" "2026" "improvement"* and *"tree tensor" "approximation guarantee" "2026"*. No universal improvement or matching complexity lower bound was located. Randomized methods and better practical error estimates require a separate comparison with the displayed worst-case statement.